

Maximum likelihood estimation (MLE) recap

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- Probability distribution models are most commonly fit by **maximum likelihood estimation** (MLE).

Gaussian (normal) distribution

- Consider the Gaussian distribution, given by a density function

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

where $\mu \in \mathbb{R}, \sigma^2 \in \mathbb{R}_+$ are **parameters**.

- We have some data e.g., measurements of physical height of people in some population. E.g.,

$$X = \{161cm, \}$$